MIL-C-16385E(SHIPS)
17 November 1972
SUPERSEDING
MIL-C-16385D(SHIPS)
28 July 1964
(See 6.5)

MILITARY SPECIFICATION

CARTRIDGE, CARBON DIOXIDE, INFLATABLE LIFE PRESERVER USE

1. SCOPE

1.1 This specification covers carbon dioxide cartridges for use with Underwater Demolition Team and Yoke Type SCUBA CO_2 Inflatable Life Preservers (see 6.1).

1.2 Classification. The cartridges furnished under this specification shall be of the following types, as specified (see 6.1 and 6.2).

Type I - Cartridges containing 17 to 21 grams of CO_2

Type II - Cartridges containing 30 to 34 grams of CO_2

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-I-558 - Ink, Marking Stencil, Opaque, For Nonporous Surfaces (Metals, Glass, Etc.).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-676 - Boxes, Set-Up, Paperboard.
PPP-T-97 - Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-M-19595 - Magnetic Effect Limits for Nonmagnetic Equipment Used In The Proximity of Magnetic Influence Ordnance.

STANDARDS

FEDERAL

FED-STD-151 - Metals; Test Method.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Drive, Chicago, Illinois 60606.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production, samples shall be tested as specified in 4.2 (see 6.4).

3.2 Magnetic effect. The cartridges shall conform to the magnetic effect limits of MIL-M-19595.

3.3 Corrosion resistance. The cartridges shall be rendered corrosion resistant by anodization passivation or other applicable treatment. The cartridge shall be capable of withstanding a 168-hour synthetic sea water spray test (see 4.7.4).

3.4 Filling. The maximum filling density of the cartridge shall be 60 percent (percent ratio of the weight of gas in a container to the weight of water that the container will hold).

3.5 Gas charge.

3.5.1 Type I. Type I filled cartridges shall contain not less than 17 and not more than 21 grams of CO_2.

3.5.2 Type II. Type II filled cartridges shall contain not less than 30 and not more than 34 grams of CO_2.

3.6 Dimensions. The dimensions of the cartridges shall be as shown on figure 1, except that type II cartridges may be increased a maximum of 1 inch in length.

3.7 Puncturable area. The puncturable area shall be a coined area that is an integral part of the cartridge wall or shall be a puncturable sealing cap securely welded to the cartridge.

3.8 Leakage. Filled cartridges shall show no visible leakage when subjected to the test specified in 4.7.1.

3.9 Hydrostatic pressure. Cartridge shall not burst when subjected to the hydrostatic pressure test specified in 4.7.3.

3.10 Identification. Each cartridge shall be marked with ink conforming to TT-I-558, black or dark blue in color, and the marking area shall be coated with a compatible, clear lacquer. The markings shall consist of legible and durable numerals and letters not less than 1/8-inch high and shall consist of the following specific information:

US

Nonmagnetic "Ψ" (low mu)

Specification No. MIL-C-16385E
Type No. (I or II) (for UDT or SCUBA as applicable)
Registered symbol of manufacture
Tare weight (grams)
Indicated minimum gross weight (see 3.10.1)
Contract or order No.

3.10.1 Indicated minimum gross weight shall be the weight of the cartridge plus the weight of the minimum gas charge.

3.11 Workmanship. The carbon dioxide filled cartridges shall be uniform in quality, smooth, clean, free from waviness, splits, cracks (fractures), nicks, burrs, dents, pit marks, sharp edges, embedded foreign matter or other irregularities or defects which could effect performance, reliability, or durability. The carbon dioxide filled cartridges shall conform to the quality and grade of product established by this specification. The occurence of defects shall not exceed the acceptance criteria established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the

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inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests. The inspection and testing of the cartridges shall be classified as follows:

(a) First article inspection (see 4.2).
(b) Quality conformance inspection (see 4.5).

4.2 First article inspection. Three sample cartridges shall be subjected to the visual and dimensional examination of 4.5.2 and the tests of 4.7.

4.2.1 First article inspection report. The contractor shall submit to the procuring activity, for approval, a complete report of all the first article tests conducted.

4.3 Samples for magnetic test. The contractor shall submit three samples of uncharged fabricated cartridges to the command or agency concerned (see 6.3) for the magnetic effect test specified in 4.7.5. Samples shall be produced by the same process and at the same plant location proposed by the contractor for execution of the contract or order. Samples shall be plainly identified by securely attached durable tags marked with the following information:

Sample for magnetic permeability test
Name of product
Name of manufacturer
Contract or order number

4.4 Sampling for quality conformance inspection.

4.4.1 Lot. All cartridges of the same type presented at one time shall be considered a lot for purposes of quality conformance inspection.

4.4.2 Sampling for examination. A random sample of cartridges shall be selected from each lot in accordance with inspection level II of MIL-STD-105 for the examination specified in 4.5.2. Lot acceptance shall be based on an AQL of 1.0 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.3 Sampling for tests. A random sample of cartridges shall be selected from each lot in accordance with the inspection levels shown in table I as specified in MIL-STD-105 for the tests specified:

Table I - Sampling for tests.

Tests	Applicable paragraph	Sample size	Acceptance criteria
Dimensional inspection (gage)	4.7.2	Inspection Level II	0.015
Hydrostatic pressure	4.7.3	Inspection Level S-3	0.015
Magnetic effect	4.7.5	6	0

4.5 Quality conformance inspection.

4.5.1 Production tests. Each completed cartridge shall be subjected to the gas charge and leakage test (see 4.7.1).

4.5.2 Examination. Sample cartridges selected in accordance with 4.4.2 shall be visually and dimensionally examined to verify conformance to the requirements of this specification.

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4.5.3 Classification of defects. Inspection of the cartridges shall be conducted as specified in table II:

Table II - Classification of defects.

Critical	None.
Major	
101	Material not as specified.
102	Cartridge shape (or form) not uniform, not concentric.
103	Neck axis not in alinement with body axis.
104	Surface rough, or contains waviness, splits, cracks (fractures), nicks, burrs, dents, pit marks, sharp edges, embedded foreign matter or otherwise defective.
105	Threads stripped, torn, broken, or damaged.
106	Sealing cap (if used) not securely welded to cartridge.
Minor	
201	Overall length exceeds specified value.
202	Overall diameter not as specified.
203	Identification marking missing, incorrect or illegible.

4.6 Test conditions.

4.6.1 Atmospheric conditions. Unless otherwise specified (see 6.1), all tests required by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury and a temperature of 77 ± 18 degrees Fahrenheit (°F.).

4.7 Test methods.

4.7.1 Gas charge and leakage. The cartridges shall show no visible leakage when tested as follows:

(a) The weight of the empty cartridge shall be recorded. Filled cartridges shall be weighed to determine compliance with 3.10.1 and shall be immersed in water heated to 190° to 200°F. Cartridges shall remain in the water for 30 minutes minimum after temperature stabilization. All cartridges which show visible leakage shall be removed. The gas charge shall be verified by weighing each cartridge after liquid dip cleaning.

4.7.2 Dimensional inspection (gage). Filled cartridges shall be dimensionally checked with a "go" gage conforming to the cavity dimensions shown on figure 2. Failure of punctur-able area of cartridge to seat firmly against bottom of gage cavity shall be cause for rejection.

4.7.3 Hydrostatic pressure test. Cartridges shall be subjected to an internal hydrostatic pressure of 6500 pounds per square inch gage (p.s.i.g) without bursting.

4.7.4 Synthetic sea-water spray test. The cartridges shall be subjected to the synthetic sea-water test, method 812 of Federal Test Method Standard 151 to determine conformance with 3.3.

4.7.5 Magnetic effect. The cartridges shall be subjected to the magnetic permeability test of MIL-M-19595.

4.8 Inspection of preparation for delivery. The packaging, packing, and marking of cartridges shall be inspected to determine conformance with section 5 herein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Packaging.

5.1.1 Level A. Twelve cartridges shall be packaged in a paperboard folding or set-up box conforming to PPP-B-566 or PPP-B-676, respectively; at the option of the contractor, cartridges shall be separated by paperboard cell partitions. Box closure shall be as specified in the applicable box specification.

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5.1.1.1 Intermediate containers. Eighteen unit packages as specified in 5.1.1 shall be intermediately packaged in fiberboard boxes conforming to PPP-B-636, class weather-resistant. Box closure shall be as specified in the appendix to PPP-B-636.

5.1.2 Level C. Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use or onboard storage for early installation. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2 Packing.

5.2.1 Level A. Unless otherwise specified, cartridges, packaged as specified (see 6.2), shall be packed in containers conforming to any one of the following specifications at the option of the contractor:

PPP-B-576	Class 2
PPP-B-585	Class 3
PPP-B-591	Class 2
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-636	Weather-resistant

Shipping containers shall be closed and strapped in accordance with the applicable box specification or appendix thereto, except fiberboard boxes shall be banded with tape conforming to type III of PPP-T-97, and the appendix thereto. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification. Intermediate fiberboard boxes, closed, sealed, and banded as specified herein, may be used as the shipping container.

5.2.2 Level B. Cartridges, packaged as specified (see 6.2) shall be packed in containers conforming to any one of the following specifications at the option of the contractor:

PPP-B-576	Class 1
PPP-B-585	Class 1 or 2
PPP-B-591	Class 1
PPP-B-601	Domestic type
PPP-B-621	Class 1
PPP-B-636	Domestic

Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood cleated boxes shall not exceed the weight limitations of the applicable box specification. Intermediate fiberboard boxes, closed, sealed, and banded as specified herein may be used as the shipping container.

5.2.3 Level C. Cartridges, packaged as specified (see 6.2) shall be packed in containers which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the first receiving activity for immediate use or onboard storage for early installation. This level in general shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may be the supplier's commercial practice when such meets the requirements of this level.

5.3 Marking. In addition to any special markings required (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129 and as follows:

5.3.1 Each interior or exterior container shall be durably and legibly marked with the following information in such a manner that the markings shall not be damaged when containers are opened;

⇓ (low Mu)
Type I - Cartridges, 17 to 21 gram, CO_2 filled (for Underwater Demolition Team CO-2 Inflatable Life Preservers)

or (as applicable)

Type II - Cartridges, 30 to 34 gram, CO_2 filled (for Life Preserver, Yoke Type, SCUBA, U. S. Navy Mark 3)
Specification number
Date of manufacture (month and year).

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5.3.2 Unit containers. Marking ink, paint, or other material containing ferrous oxide, or substances that do not conform to MIL-M-19595 shall not be used in unit containers.

5.4 Use of polystyrene (loose-fill) material.

5.4.1 For Level A packaging and Level A and B packing. Use of polystyrene (loose-fill) material is prohibited for Level A packaging and Level A and B packing applications such as cushioning, filler, and dunnage.

5.4.2 For domestic shipment and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.2) use of polystyrene (loose-fill) material for domestic shipment and level C packaging and packing applications such as cushioning, filler, and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked as follows:

CAUTION

Contents cushioned, etc., with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard, or transparent flexible cellular material.

6. NOTES

6.1 Intended use. Cartridges furnished under this specification are intended for use in the following Life Preservers:

Type I - UDT life preservers conforming to MIL-L-16383.
Type II - Yoke type SCUBA life preservers conforming to MIL-L-24047.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Types (see 1.2).
(c) Atmospheric condition, if different (see 4.6.1).
(d) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).
(e) Special marking, other than as specified (see 5.3).
(f) Use of polystyrene material (see 5.4.2).

6.3 Applications for submitting material or equipment, or both, for magnetic effect tests should be made to: Officer-in-Charge, U.S. Naval Weapons Laboratory, Dahlgren, Virginia. The application should be made as far in advance of the anticipated submission date as is practicable, and should contain the following information:

(a) Title and type (see 1.2).
(b) Size and type of unit containers.
(c) Quantity submitted.
(d) Date of delivery.
(e) Date completion is desired.
(f) Reports required.
(g) Disposition instructions.

6.3.1 Shipments of material or equipment submitted for magnetic effect test are to be addressed to:

Receiving Officer
U. S. Naval Weapons Laboratory
Dahlgren, Virginia

6.3.1.1 The exterior containers of the shipments should be marked as follows:

For inspection only
Contract identification
(Agency and contract or order number)

6.4 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

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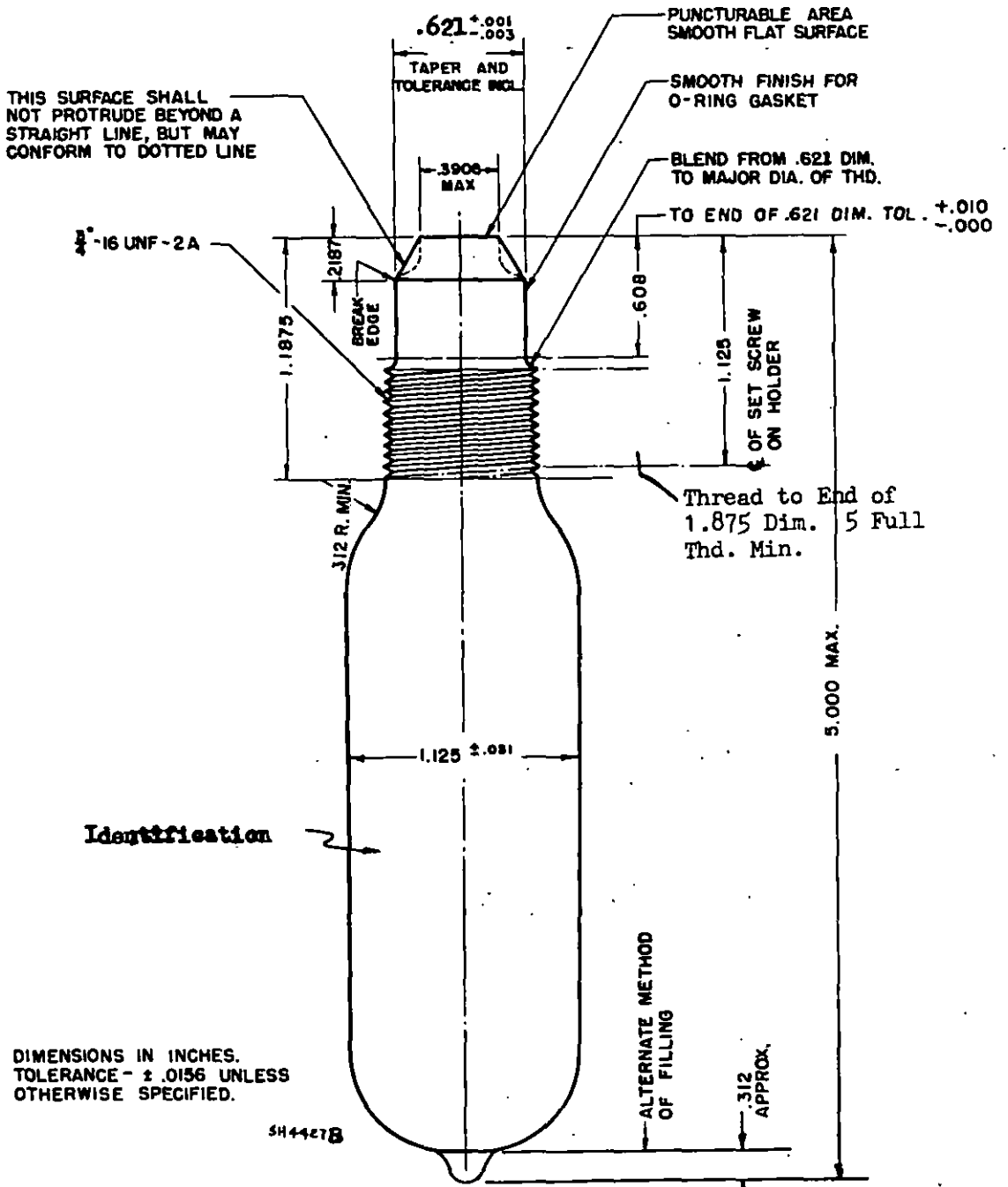
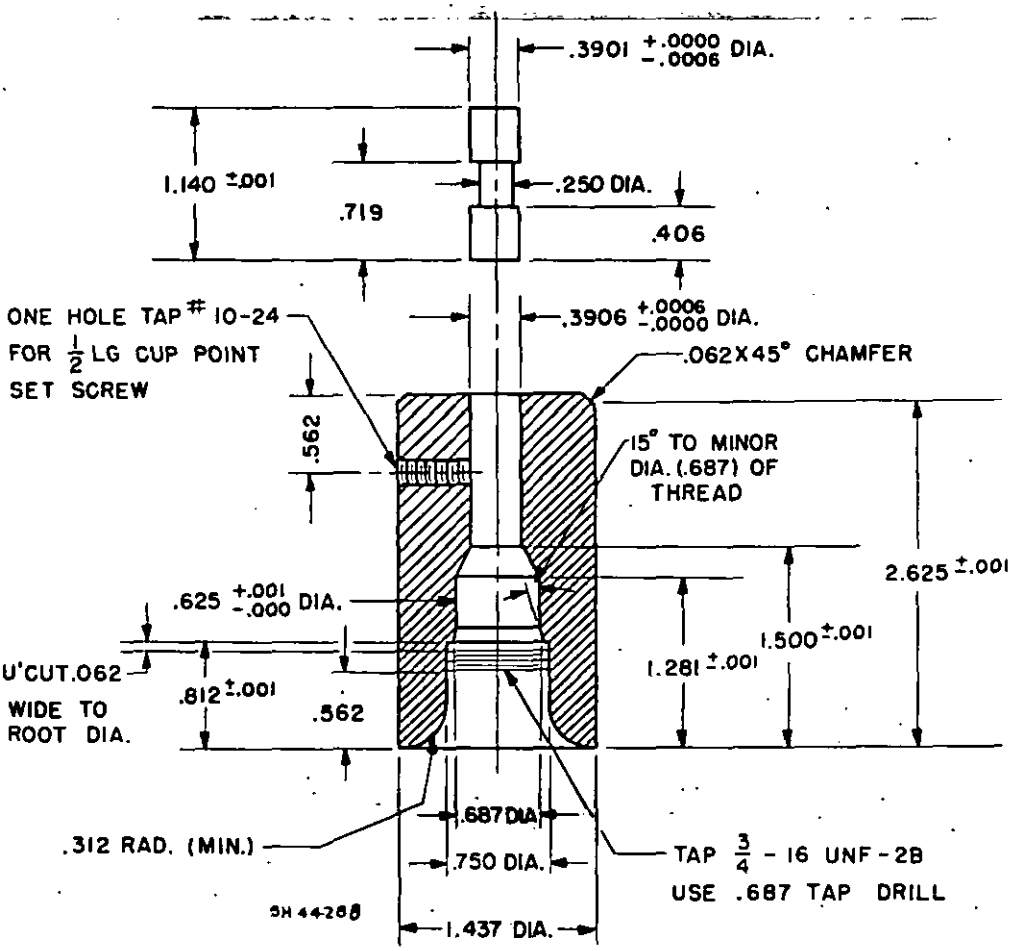


Figure 1 - Dimensions of cartridge.

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TOLERANCE ±.0156 UNLESS
OTHERWISE SPECIFIED.

Figure 2 - Cavity dimensions.

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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